General and Structural Chemistry

Practice Assignments-2

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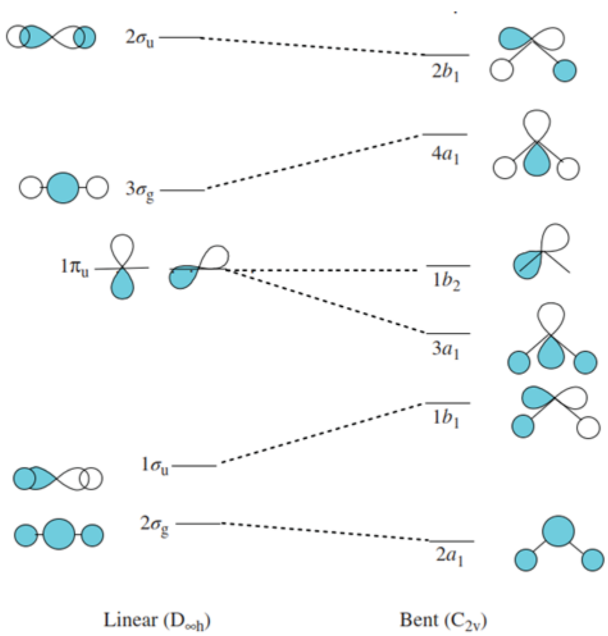
Q1. In the following table some bond energy and bond length values are given. How can you justify the bond energy trends found in a given group using the MOT concepts?

Bond	H-C	H-Si	H-Ge	H-Sn	H-N	H-P	H-As	H-O	H-S	H-Se	H-Te
Bond Energy (kJ/mol)	411	318	288	251	386	322	247	459	363	276	238
Bond Length (pm)	109	148	153	170	101	144	152	96	134	146	170

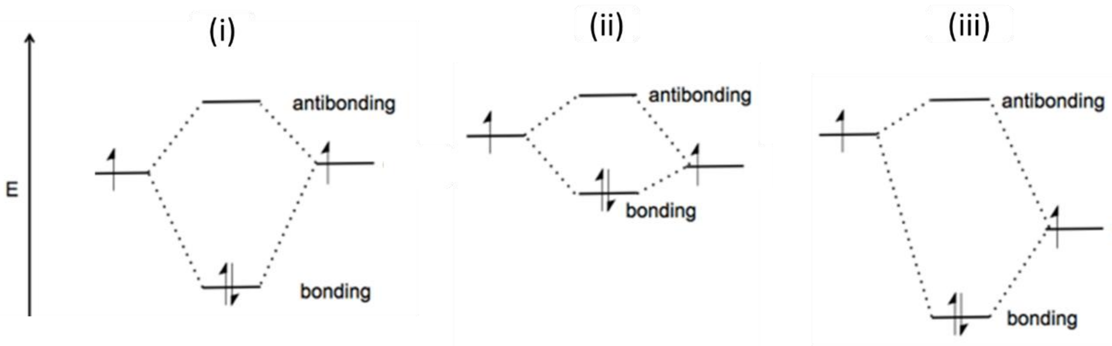
Q2. Walsh diagram is given below. The energy increases from bottom to top. Predict the molecular geometries for the ground and first electronic excited state configuration for BeH_2? Justify your answers. What are the HOMOs for the ground and first excited states, respectively?

Walsh Diagram:



Q3. By using IR spectroscopy, bond vibration frequencies can be determined. The stronger the bond, the higher the vibration frequency. IR frequencies (ν(CO) stretching frequencies) of CO(g) and CO in $Cr(CO)_6$ are 2140 cm^{-1} and 2000 cm^{-1} respectively. Justify your answer.

Q4. The following schematic MO diagrams (i, ii, & iii) depict the C-C, C-Cl, and C-Br bond formations. Match each diagram to the correct bond formation. Explain.



Q5. True or False regarding s-p mixing:

(i) There is ***no significant*** 2s–2p orbital mixing for O_2 and F_2.

(ii) Period 2 homonuclear diatomic molecules, there is a change of energy order of the $1\pi_u$ ($\pi_{2px} = \pi_{2py}$) and $2\sigma_g$ (σ_{2pz}) orbitals starting from O_2.

(iii) *Evidence that s-p mixing occurs* is supported by the *photoelectron spectroscopy* (*PES*).

Q6. If 6 half-filled p-orbitals combine to form pi-MOs (pi-molecular orbitals), how many pi-MOs will be formed? How many nodes are there in the respective frontier orbitals (HOMO and LUMO)?

Q7. What is SOMO? Name a chemical species which has SOMO.

Q8. Mention two diatomic species with bond order 1 and 2 respectively which have only pi bond(s) and no sigma bond.

Q9. What are reactive oxygen species (ROS)? Give an example. Use MOT to show why they are reactive.

Q10. Draw a Molecular Orbital Energy-Level Diagram of HCl. How many molecular orbitals are there? What is the bond order? Comment on the following: (i) contributing coefficients of atomic orbitals towards different MOs and (ii) location and nature of bonding and unshared electron pairs. How are HS^- MOs different from that of HCl?

Q11. For heteronuclear diatomic molecules, show that

$$E_{\pm} = \frac{(\alpha_A + \alpha_B)}{2} \pm \frac{1}{2}(\alpha_A - \alpha_B)\left\{1 + \left(\frac{2\beta}{\alpha_A - \alpha_B}\right)^2\right\}^{1/2}$$

From here show that when the energy difference $|\alpha_A - \alpha_B|$ is very large, the energies of the resulting molecular orbitals differ only slightly from those of the atomic orbitals, which implies in turn that the bonding and antibonding effects are small. That is: "The *strongest bonding* and *antibonding effects* are obtained when the two contributing orbitals have *similar energies*."

Q12. For a heteronuclear diatomic with $S \approx 0$,

$$c_A = \frac{1}{\sqrt{1 + \left(\frac{\alpha_A - E}{\beta}\right)^2}} = \frac{\beta}{\sqrt{\beta^2 + (\alpha_A - E)^2}}$$

$$c_B = -\left(\frac{\alpha_A - E}{\beta}\right) c_A = \frac{\alpha_A - E}{\sqrt{\beta^2 + (\alpha_A - E)^2}}$$

Calculate the c_A and c_B for bonding and antibonding MOs of HCl.

Q13. BN (with eight valence electrons) is a paramagnetic molecule. Justify it.

Q14. What would be the MOs if H_2O were linear in shape?

Q15. H2O has a bond angle of 104.5°. Calculate the p-character of the O-H bonds. Calculate the s-character of the oxygen lone pairs.

Q16. Mention three major factors on which the strength of orbital interactions depends.

Q17. (i) What is the bond order of H_2O? There are **two O-H bonds**, one on either side of H_2O. (ii) Is H_2O paramagnetic or diamagnetic? (iii) How does the MO diagram of H_2O suggest its bent shape and structure?

Q18. Write the normalized hybrid wavefunctions for sp^3 hybridization. Take any two hybrids and show that they are orthogonal to each other.

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